Sean Yépez · ME344 Summer 2026 · Final Project, Option 2

↗ github.com/seanyepez/me344ricechem

Problem: grading consumes the time meant for teaching

I love teaching, but for every 8 hours I spend teaching, 7 go to grading. I want to invert that ratio. Treemarks is my AI grading system for STEM classes: every decision checked against a rubric tied to the underlying logic, physics, and math.

AI checks every rubric decision → the TA reviews uncertain cases → students receive feedback sooner

Sean Yépez · Dr. Mourad Bouache · Google

RICECHEM · THE SUPERVISED BENCHMARK

RiceChem is the benchmark recent papers use: a chemistry exam graded against instructor rubrics. 1,264 responses · 4 questions · 27 rubric items → 8,392 TA-labeled TRUE/FALSE decisions. Agreement is measured against the TA's labeling.

- Sonkar et al. (2024) · introduces RiceChem and rubric-based grading of long-form answers
- Rao & Callison-Burch (2026) · Autorubric, rubric-based LLM evaluation
- Ferrer et al. (2026) · when to trust LLM graders

RESEARCH QUESTIONS

RQ1 · Quality: can tuning make a local model accurate enough for rubric grading?

RQ2 · Systems: how do CPU, GPU, and TPU change latency, throughput, and utilization?

RQ3 · Operations: which configuration meets the quality bar in under 24 hours at lowest cost?

PROPOSAL · SYSTEM + EXPERIMENT

Containerize and run a supervised experiment on three compute lanes

ORCHESTRATE

Docker + GKE Job

pinned image · reusable K8s manifest

STORAGE + I/O

GCS + range cache

authorized data · checkpoints · aggregate receipts

TRAIN + SERVE

CPU + A100
PyTorch / QLoRA

CUDA · vLLM dynamic batching

TPU v5e-8
JAX / XLA

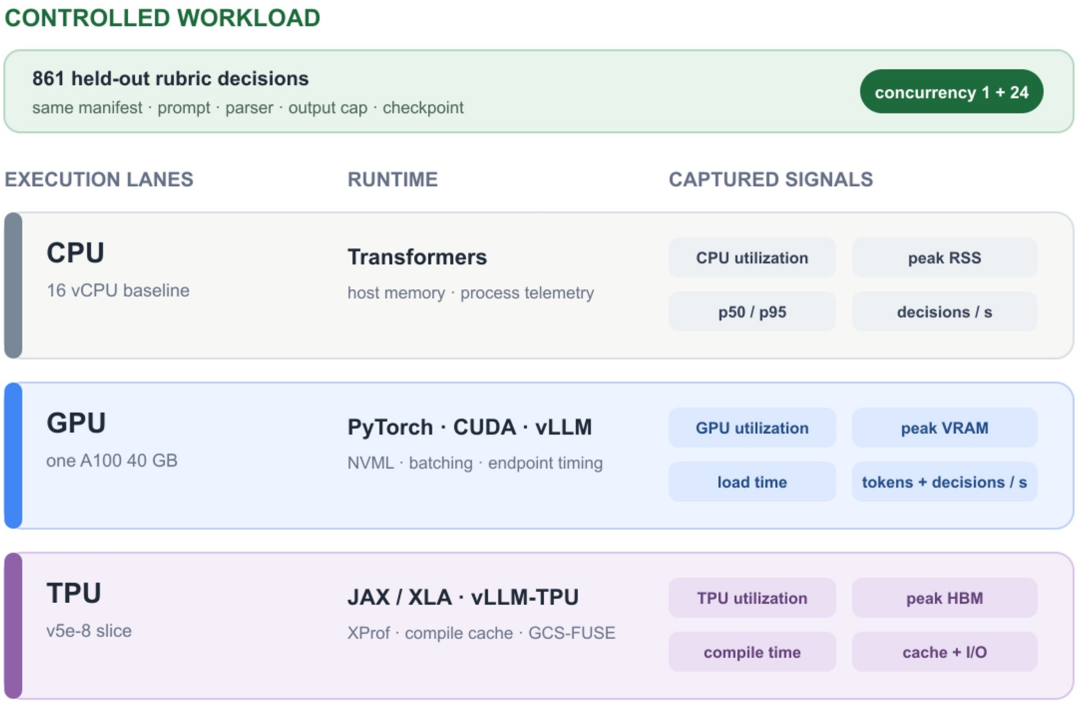
compiled train / serve step

EXPERIMENT DESIGN · 8,392 TA-LABELED DECISIONS

TRAIN · 6,700 · 80% update adapters	VALIDATE 831 · 10%	TEST 861 · 10%

Validation picks the checkpoint. The test set is opened once, at the end.

Measurements across three kinds of compute



WORKLOAD
861 rubric decisions from the RiceChem test split · c=1 and c=24

COMPUTE
wall time · p50/p95 · decisions/s · wall/step time as cycle proxy

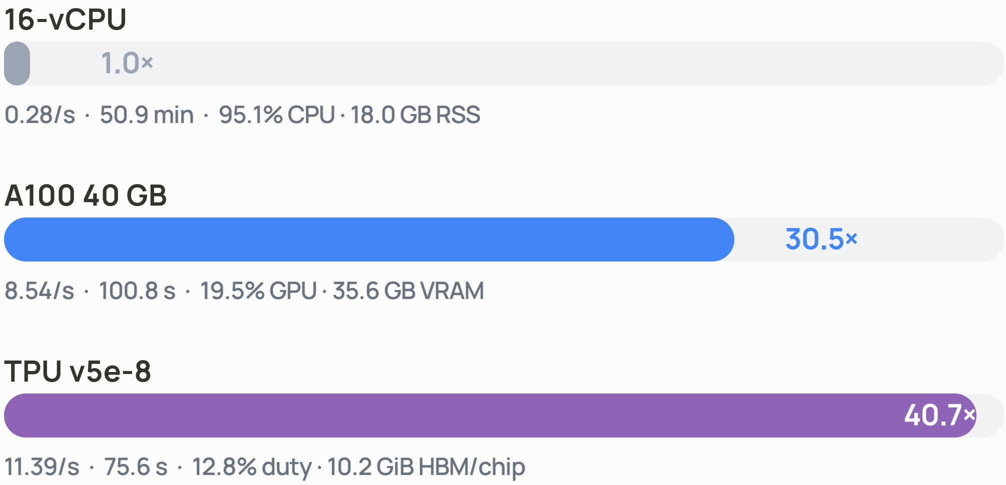
MEMORY
CPU mCPU + RSS · A100 NVML + VRAM · TPU host memory only

INPUT / I/O
GCS range cache fixed startup; load and XLA compile time not isolated

RESULTS · CONTROLLED 4B SERVING PROFILE

Accelerators cut single-stream grading time 30–41×

THROUGHPUT SPEEDUP · CONCURRENCY 1 · CPU = 1×



CONCURRENT SERVING · c = 24
A100 123.9/s
$0.007 per 861 decisions
TPU 98.4/s
$0.023 per 861 decisions

GEMMA 4B QUALITY
The 4B agrees only ≈72% of the time, below the grading bar.
Speed alone does not make it deployable.

Same 861-decision workload and configured merged-checkpoint path. CPU c=24 did not complete under memory/tunnel pressure. Costs are active-time list-price proxies; TPU device telemetry captured with tpu-info (libtpu).

CONCLUSION

What ME344 changed for AI grading at Stanford Civil Engineering

From local prototype to cloud-hosted. At the start of the course, grading ran on a student laptop. Now inference runs in a hosted Google Cloud solution, with controlled ML experiments testing subject-specific models.

Throughput closes the feedback loop. Without local-LLM limits and with room to scale, grades that typically take a TA a week or more could come back as fast as the next morning.

Quality gates model usage. Smaller models do not offer the accuracy grading needs: at roughly 72% agreement, TAs would re-review too much. The fine-tuned 27B reached 83%, competitive with frontier models at about 1/9 the serving cost.

The operating rule. Among graders whose accuracy and throughput meet the standard, choose the cheapest.

NEXT Serve the 27B on TPU · test transfer to unseen questions · estimate the economics of training custom models versus frontier inference.